Upgrade PSP database from 11g to 12c using Goldengate

- 1. Setup GG on source and target db server
  - 1.1 Setup environment on the source and target db server:
  - 1.2 Install and configure Goldengate on the source and target db server
    - 1.2.1 Install 12c Goldengate for 11g database on the source and target db server
    - 1.2.2 Configure the database for Goldengate on the source db server
    - 1.2.3 Configure Goldengate on the source db server
  - 1.3 Create and start extract and pump from source to target
  - 1.4 Initial Sync
    - 1.4.1 Clone/restore source db on new/target (using RMAN or with the help of storage admin)
    - 1.4.2 Sanity check
  - 1.5 Final step for Goldengate replication from source(11g) to target (11g)
  - 1.6 Setup Veridata
  - 1.7 Upgrade Oracle database on target to 12c
  - 1.8 Setup Goldengate from 11g to 12c on new-box
    - 1.8.1 set up GG for 12c on new/target db server
    - 1.8.2 Set up and start integrated replicat on 12c database
  - 1.9 Setup Goldengate replication from 12c to 11g
  - 1.10 Setup GG monitoring on source and target db server
- 2. Cutover to 12c
  - 2.1 Check OOS (Out-Of-Sync)
  - 2.2 shut down application on current 11g
  - 2.3 Lock application user accounts on current database (11g)
  - 2.4 Verify GG replication
  - 2.5 Unlock application user accounts on current database (12c)
  - 2.5 start application on new database 12c

1. Setup GG on source and target db server

1.1 Setup environment on the source and target db server:

- Create the directories on the source and target db host

```
. /l/<ORACLE_SID> # Need to replace <ORACLE_SID> with real name

sudo chown -R oracle:dba /u01/ogg
mkdir -p /u01/ogg/scripts
cd /u01/ogg/
mkdir -p $ORACLE_SID/scripts
mkdir -p $ORACLE_SID/export
mkdir -p $ORACLE_SID/logs
mkdir -p $ORACLE_SID/goldengate/binaries
mkdir -p $ORACLE_SID/goldengate/scripts
mkdir -p $ORACLE_SID/goldengate/ogg_monitoring
```

- Copy scripts – **Taken care**

```
cd /u01/ogg/$ORACLE_SID/goldengate/scripts
copy scripts under this location from perf/sys db server

cd /u01/ogg/scripts
copy scripts under this location from perf/sys db server
```

- You should have the following files /u01/ogg/$ORACLE_SID/goldengate/scripts:

```
oracle@pprfpspdb800.ie.intuit.net:/u01/ogg->ls pspqpf01/goldengate/scripts/
GG01_install_goldengate.log  GG01_install_goldengate.sh  GG02_configure_db_for_goldengate.sh
GG03_configure_goldengate.sh
```

- You should have the following files /u01/ogg/scripts:

```
oracle@pprfpspdb800.ie.intuit.net:/u01/ogg->ls /u01/ogg/scripts/
iac_run_defgen.sh  monitor_ggserr.sh  monitor_gg_sync_ihp.sh  monitor_gg_sync.sh  monitor_gg_sync_table_ihp.
sql  monitor_pqboc03.cfg  rotate_ggserr.sh
```

1.2 Install and configure Goldengate on the source and target db server

1.2.1 Install 12c Goldengate for 11g database on the source and target db server

```
. /l/<Oracle sid name>
cd /u01/ogg/$ORACLE_SID/goldengate/scripts
./GG01_install_goldengate_11g.sh |tee GG01_install_goldengate.log
```

- Run environment file again since the script above updated the file and create direcotry for trail files

```
. /l/$ORACLE_SID

mkdir $GG_HOME/dirdat/<source_sid>
mkdir $GG_HOME/dirdat/<target_sid>
```

1.2.2 Configure the database for Goldengate on the source db server

```
./GG02_configure_db_for_goldengate_11g.sh <ggs/ggt password> |tee GG02_configure_db_for_goldengate.log
```

Note:

1. You will get the prompt "Is the primary db in QDC? (Yes/No)?" and another prompt "Please enter the database sys password".
2. Check the error. The following errors can be ignored

```
egrep "ORA-|SP2-" GG02_configure_db_for_goldengate.log
ORA-12920: database is already in force logging mode
ORA-02379: profile GENERIC_APP_PROFILE already exists

Any error/warning about enable_goldengate_replication parameter if it is running against 11g database.
```

1.2.3 Configure Goldengate on the source db server

WARNING : Goldengate command "add schematrandata <schema name>" could block application transactions for a few minutes, you should

1. **Run the following command only night PST hours 8 PM - 11 PM**
2. **Login the database with SQLPlus or other tool and constantly check # of active connections while running the following command. Kill the blocker session (owned by "ggs") if the # is over 50.**

```
 cd $GG_HOME
./ggsci
add CREDENTIALSTORE
ALTER CREDENTIALSTORE ADD USER ggs password <ggs password> alias ggsource
ALTER CREDENTIALSTORE ADD USER ggt password <ggt password> alias ggtarget
ALTER CREDENTIALSTORE ADD USER ggs@<lvdc_db> password <ggs password> alias ggsource_remote
ALTER CREDENTIALSTORE ADD USER ggt@<lvdc_db> password <ggt password> alias ggtarget_remote
info CREDENTIALSTORE
dblogin USERIDALIAS ggsource
dblogin USERIDALIAS ggtarget
dblogin USERIDALIAS ggsource_remote
dblogin USERIDALIAS ggtarget_remote
exit
echo "dblogin USERIDALIAS ggsource" > diroby/dblogin_s.oby
echo "dblogin USERIDALIAS ggtarget" > diroby/dblogin_t.oby
echo "dblogin USERIDALIAS ggsource_remote" > diroby/dblogin_s_remote.oby
echo "dblogin USERIDALIAS ggtarget_remote" > diroby/dblogin_t_remote.oby
obey ./diroby/dblogin_s.oby
obey ./diroby/dblogin_t.oby
obey ./diroby/dblogin_s_remote.oby
obey ./diroby/dblogin_t_remote.oby
```

- Create wallet for encryption on source db server

```
cd $GG_HOME
./ggsci
GGSCI (pprfpspdb800.ie.intuit.net) 1> create wallet

Created wallet at location 'dirwlt'.

Opened wallet at location 'dirwlt'.

GGSCI (pprfpspdb800.ie.intuit.net) 2> add masterkey

Master key 'OGG_DEFAULT_MASTERKEY' added to wallet at location 'dirwlt'.

GGSCI (pprfpspdb800.ie.intuit.net) 3> info masterkey
Masterkey Name:                 OGG_DEFAULT_MASTERKEY
Creation Date:                  Wed Nov 22 22:16:24 2017

Version:        Creation Date:                  Status:
1               Wed Nov 22 22:16:24 2017        Current

GGSCI (pprfpspdb800.ie.intuit.net) 4>

GGSCI (pprfpspdb800.ie.intuit.net) 4>  info masterkey version 1
Masterkey Name:                 OGG_DEFAULT_MASTERKEY
Creation Date:                  Wed Nov 22 22:16:24 2017
Version:                        1
Renew Date:                     Wed Nov 22 22:16:24 2017
Status:                         Current
Key Hash (SHA1):                0x145539DA265F70D2562947409EDCEA98BA4665A2

GGSCI (pprfpspdb800.ie.intuit.net) 5> exit

Make a note of the above key hash value
The wallet created a file under sub-directory dirwlt
ls -l ./dirwlt
total 4
-rw-r----- 1 oracle oinstall 685 Mar 19 11:49 cwallet.sso
```

- Copy wallet and credential store files to target db server at similar location

```
. /l/pspuat
cd $GG_HOME
oracle@pprfpspdb800.ie.intuit.net:/u01/ogg/pspqpf01/12.1.2.1.6_11g/dirwlt->scp cwallet.sso oracle@pprfpspdb900.
ie.intuit.net:/u01/ogg/psplpf01/12.1.2.1.6_11g/dirwlt/

********************************************************************************
This is a private computer system containing information that is proprietary
and confidential to the owner of the system.  only individuals or entities
authorized by the owner of the system are allowed to access or use the system.
Any unauthorized access or use of the system or information is strictly
prohibited.

All violators will be prosecuted to the fullest extent permitted by law.
********************************************************************************

cwallet.
sso
100%  685     0.7KB/s   00:00
oracle@pprfpspdb800.ie.intuit.net:/u01/ogg/pspqpf01/12.1.2.1.6_11g/dirwlt->cd ../dircrd/
oracle@pprfpspdb800.ie.intuit.net:/u01/ogg/pspqpf01/12.1.2.1.6_11g/dircrd->ls -ltr
total 8
-rw-r----- 1 oracle dba 693 Nov 22 22:11 cwallet.sso
oracle@pprfpspdb800.ie.intuit.net:/u01/ogg/pspqpf01/12.1.2.1.6_11g/dircrd->scp cwallet.sso oracle@pprfpspdb900.
ie.intuit.net:/u01/ogg/psplpf01/12.1.2.1.6_11g/dircrd/

********************************************************************************
This is a private computer system containing information that is proprietary
and confidential to the owner of the system.  only individuals or entities
authorized by the owner of the system are allowed to access or use the system.
Any unauthorized access or use of the system or information is strictly
prohibited.

All violators will be prosecuted to the fullest extent permitted by law.
********************************************************************************

cwallet.
sso
100%  693     0.7KB/s   00:00
```

- Verfiy wallet on target

```
. /l/pspuat
cd $GG_HOME
./ggsci
GGSCI (pprdqbodb322.ie.intuit.net) 1> open WALLET
Opened wallet at location 'dirwlt'.
GGSCI (pprdqbodb322.ie.intuit.net) 2> info masterkey version 1
Masterkey Name:                 OGG_DEFAULT_MASTERKEY
Creation Date:                  Wed Feb  4 13:00:02 2015
Version:                        1
Renew Date:                     Wed Feb  4 13:00:02 2015
Status:                         Current
Key Hash (SHA1):                0x18834E55EF983C46BABF46DF9200E2CFF6B22CE3

NOTE: make sure the hash keys must be identical on source and target

GGSCI (pprdqbodb322.ie.intuit.net) 3> info  CREDENTIALSTORE
Reading from ./dircrd/:
Default domain: OracleGoldenGate
For source

  Alias: ggtarget
  Userid: ggt

  Alias: ggtarget
  Userid: ggt
```

- Create an obey file for login on target db server and verify login

```
cd $GG_HOME
echo "dblogin USERIDALIAS ggsource" > diroby/dblogin_s.oby
echo "dblogin USERIDALIAS ggtarget" > diroby/dblogin_t.oby
echo "dblogin USERIDALIAS ggsource_remote" > diroby/dblogin_s_remote.oby
echo "dblogin USERIDALIAS ggtarget_remote" > diroby/dblogin_t_remote.oby

verify login
./ggsci
obey ./diroby/dblogin_s.oby
obey ./diroby/dblogin_t.oby
obey ./diroby/dblogin_s_remote.oby
obey ./diroby/dblogin_t_remote.oby
exit
```

- DDL setup on on **source** db as it is on 11g version. you will be prompted for name of schema. The answer is GGS

```
cd $GG_HOME
sqlplus / as sysdba
1.
SQL> @marker_setup.sql
Marker setup script

You will be prompted for the name of a schema for the Oracle GoldenGate database objects.
NOTE: The schema must be created prior to running this script.
NOTE: Stop all DDL replication before starting this installation.

Enter Oracle GoldenGate schema name:GGS


Marker setup table script complete, running verification script...
Please enter the name of a schema for the GoldenGate database objects:
Setting schema name to GGS

MARKER TABLE
-------------------------------
```

```
OK

MARKER SEQUENCE
-------------------------------
OK

Script complete.

2.
SQL> @ddl_setup.sql
Oracle GoldenGate DDL Replication setup script

Verifying that current user has privileges to install DDL Replication...

You will be prompted for the name of a schema for the Oracle GoldenGate database objects.
NOTE: For an Oracle 10g source, the system recycle bin must be disabled. For Oracle 11g and later, it can be
enabled.
NOTE: The schema must be created prior to running this script.
NOTE: Stop all DDL replication before starting this installation.

Enter Oracle GoldenGate schema name:GGS

Working, please wait ...
Spooling to file ddl_setup_spool.txt

Checking for sessions that are holding locks on Oracle Golden Gate metadata tables ...

Check complete.


Using GGS as a Oracle GoldenGate schema name.

Working, please wait ...

DDL replication setup script complete, running verification script...
Please enter the name of a schema for the GoldenGate database objects:
Setting schema name to GGS

CLEAR_TRACE STATUS:

Line/pos                                 Error
---------------------------------------- -----------------------------------------------------------------
No errors                                No errors

CREATE_TRACE STATUS:

Line/pos                                 Error
---------------------------------------- -----------------------------------------------------------------
No errors                                No errors

TRACE_PUT_LINE STATUS:

Line/pos                                 Error
---------------------------------------- -----------------------------------------------------------------
No errors                                No errors

INITIAL_SETUP STATUS:

Line/pos                                 Error
---------------------------------------- -----------------------------------------------------------------
No errors                                No errors

DDLVERSIONSPECIFIC PACKAGE STATUS:

Line/pos                                 Error
---------------------------------------- -----------------------------------------------------------------
No errors                                No errors

DDLREPLICATION PACKAGE STATUS:

Line/pos                                 Error
```

```
---------------------------------------- ------------------------------------------------------------------
No errors                                No errors

DDLREPLICATION PACKAGE BODY STATUS:

Line/pos                                 Error
---------------------------------------- ------------------------------------------------------------------
No errors                                No errors

DDL IGNORE TABLE
-----------------------------------
OK

DDL IGNORE LOG TABLE
-----------------------------------
OK

DDLAUX  PACKAGE STATUS:

Line/pos                                 Error
---------------------------------------- ------------------------------------------------------------------
No errors                                No errors

DDLAUX PACKAGE BODY STATUS:

Line/pos                                 Error
---------------------------------------- ------------------------------------------------------------------
No errors                                No errors

SYS.DDLCTXINFO  PACKAGE STATUS:

Line/pos                                 Error
---------------------------------------- ------------------------------------------------------------------
No errors                                No errors

SYS.DDLCTXINFO  PACKAGE BODY STATUS:

Line/pos                                 Error
---------------------------------------- ------------------------------------------------------------------
No errors                                No errors

DDL HISTORY TABLE
-----------------------------------
OK

DDL HISTORY TABLE(1)
-----------------------------------
OK

DDL DUMP TABLES
-----------------------------------
OK

DDL DUMP COLUMNS
-----------------------------------
OK

DDL DUMP LOG GROUPS
-----------------------------------
OK

DDL DUMP PARTITIONS
-----------------------------------
OK

DDL DUMP PRIMARY KEYS
-----------------------------------
OK

DDL SEQUENCE
-----------------------------------
```

```
OK

GGS_TEMP_COLS
-----------------------------------
OK

GGS_TEMP_UK
-----------------------------------
OK

DDL TRIGGER CODE STATUS:

Line/pos                                 Error
---------------------------------------- -----------------------------------------------------------------
No errors                                No errors

DDL TRIGGER INSTALL STATUS
-----------------------------------
OK

DDL TRIGGER RUNNING STATUS
----------------------------------------------------------------------------------------------------------------
--------
ENABLED

STAYMETADATA IN TRIGGER
----------------------------------------------------------------------------------------------------------------
--------
OFF

DDL TRIGGER SQL TRACING
----------------------------------------------------------------------------------------------------------------
--------
0

DDL TRIGGER TRACE LEVEL
----------------------------------------------------------------------------------------------------------------
--------
0

LOCATION OF DDL TRACE FILE
----------------------------------------------------------------------------------------------------------------
--------
/u01/app/oracle/diag/rdbms/pspqpf01/pspqpf01/trace/ggs_ddl_trace.log

Analyzing installation status...


VERSION OF DDL REPLICATION
----------------------------------------------------------------------------------------------------------------
--------
OGGCORE_12.1.2.1.0_PLATFORMS_140727.2135.1

STATUS OF DDL REPLICATION
----------------------------------------------------------------------------------------------------------------
--------
SUCCESSFUL installation of DDL Replication software components

Script complete.


3.
SQL> @role_setup.sql
GGS Role setup script

This script will drop and recreate the role GGS_GGSUSER_ROLE
To use a different role name, quit this script and then edit the params.sql script to change the gg_role
parameter to the preferred name. (Do not run the script.)

You will be prompted for the name of a schema for the GoldenGate database objects.
NOTE: The schema must be created prior to running this script.
```

```
NOTE: Stop all DDL replication before starting this installation.

Enter GoldenGate schema name:GGS
Wrote file role_setup_set.txt

PL/SQL procedure successfully completed.


Role setup script complete

Grant this role to each user assigned to the Extract, GGSCI, and Manager processes, by using the following SQL
command:

GRANT GGS_GGSUSER_ROLE TO <loggedUser>

where <loggedUser> is the user assigned to the GoldenGate processes.


4.
SQL> GRANT GGS_GGSUSER_ROLE TO GGS;
Grant succeeded.

5.
Enable DDL on target
Target:
cd $GG_HOME
sqlplus / as sysdba
SQL> @ddl_enable.sql

Trigger altered.

6.
SQL> @ddl_pin GGS
PL/SQL procedure successfully completed.
PL/SQL procedure successfully completed.
PL/SQL procedure successfully completed.
SQL> exit
```

- Copy GG parameter and obey files from perf/sys environment under respective folders. Change GG process names as per prod naming convention in parameter files.

- Execute sequence.sql to support sequence replication on source database

```
SQL> @sequence.sql
Please enter the name of a schema for the GoldenGate database objects:
GGS
Setting schema name to GGS

UPDATE_SEQUENCE STATUS:

Line/pos
----------------------------------------
Error
-----------------------------------------------------------------
No errors
No errors


GETSEQFLUSH

Line/pos
----------------------------------------
Error
-----------------------------------------------------------------
No errors
No errors


SEQTRACE

Line/pos
----------------------------------------
Error
-----------------------------------------------------------------
No errors
No errors


REPLICATE_SEQUENCE STATUS:

Line/pos
----------------------------------------
Error
-----------------------------------------------------------------
No errors
No errors


STATUS OF SEQUENCE SUPPORT
--------------------------------------------------------------
SUCCESSFUL installation of Oracle Sequence Replication support
```

- execute below and update extract parameter file

```
cd $GG_HOME
login to database as sys

1.
QL> @prvtclkm.plb

Package created.


Library created.


Package body created.

2.
SQL> grant execute on sys.dbms_internal_clkm to ggs;

Grant succeeded.

3.
cd /u01/app/oracle/product/11.2.0.3/wallet

mkstore -wrl . -list
Oracle Secret Store Tool : Version 11.2.0.3.0 - Production
Copyright (c) 2004, 2011, Oracle and/or its affiliates. All rights reserved.

Enter wallet password:

Oracle Secret Store entries:
ORACLE.SECURITY.DB.ENCRYPTION.AaTDSeN74E8iv5mQLVn28GkAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
ORACLE.SECURITY.DB.ENCRYPTION.Af7bfBY/0k8Iv930IUjQEGsAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
ORACLE.SECURITY.DB.ENCRYPTION.ARCmiy8Jwk9fvwgWkVqoEhoAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
ORACLE.SECURITY.DB.ENCRYPTION.AXokytDzO091vwoe6kVd4D0AAAAAAAAAAAAAAAAAAAAAAAAAAAAA
ORACLE.SECURITY.DB.ENCRYPTION.MASTERKEY
ORACLE.SECURITY.TS.ENCRYPTION.BWnstFcEXIuVjfmT0pAAbd4CAwAAAAAAAAAAAAAAAAAAAAAAAAAA

4.
execute below command to add entry for "ORACLEGG" entry in the Oracle wallet :
mkstore -wrl . -createEntry ORACLE.SECURITY.CL.ENCRYPTION.ORACLEGG

mkstore -wrl . -createEntry ORACLE.SECURITY.CL.ENCRYPTION.ORACLEGG
        Oracle Secret Store Tool : Version 11.2.0.3.0 - Production
        Copyright (c) 2004, 2011, Oracle and/or its affiliates. All rights reserved.

        Your secret/Password is missing in the command line
        Enter your secret/Password:

        Re-enter your secret/Password:

        Enter wallet password:



5.  Re-open the wallet
SQL> ALTER SYSTEM SET ENCRYPTION WALLET CLOSE;

System altered.

SQL> ALTER SYSTEM SET ENCRYPTION WALLET OPEN IDENTIFIED BY "<password>";

System altered.

6.  repeate this steps to recycle all the redo groups.
alter system switch logfile;
```

- Grant permission to GGT

```
GRANT EXECUTE ON GGS.updateSequence to GGT;
GRANT EXECUTE ON GGS.replicateSequence to GGT;
```

1.3 Create and start extract and pump from source to target

- Start the manager on both source and target

```
cd $GG_HOME
./ggsci
start manager
info all
```

- Add supplemental logging and create extract on source db server

```
1.
cd $GG_HOME
./ggsci
dblogin USERIDALIAS ggsource
add schematrandata pspadm
obey ./diroby/create_epspuatq1.oby
info all

Note : Register extract only if db version is 12c else skip to next step
dblogin USERIDALIAS ggsource
register extract EPSPUTQ1 database

2.
generate the encrypted password for ORACLEGG
cd $GG_HOME
ggsci
ENCRYPT PASSWORD <wallet_password> BLOWFISH ENCRYPTKEY DEFAULT

GGSCI (pprfpspdb800.ie.intuit.net) 9> ENCRYPT PASSWORD <wallet_password> BLOWFISH ENCRYPTKEY DEFAULT
Using Blowfish encryption with DEFAULT key.
Encrypted password:  AACAAAAAAAAAAAMAHIYFPHGCNEQBDAOAJIYHUFEJRJNGAAJA
Algorithm used:  BLOWFISH

3. add below parameter in extract
DBOPTIONS DECRYPTPASSWORD AACAAAAAAAAAAAMAHIYFPHGCNEQBDAOAJIYHUFEJRJNGAAJA ENCRYPTKEY DEFAULT

--> start extract
obey ./diroby/dblogin_t.oby
add checkpointtable
info all
start epsputq1
sh ls ./dirdat

obey ./diroby/create_ppspuatq2.oby
start ppsputq2 on source and check trail files are getting transferred on target db server
info all
```

1.4 Initial Sync

1.4.1 Clone/restore source db on new/target (using RMAN or with the help of storage admin)

1.4.2 Lock all app user accounts

```
alter user PSPAPP account lock;
alter user PSPADM account lock;
```

1.4.2 Sanity check

- Run the following SQLs on both source and target and then compare the results

```
select owner, count(*) from dba_tables where owner in ('PSPADM') group by owner order by 1;
select index_type, count(*) from dba_indexes where owner in ('PSPADM') group by index_type order by 1;
select constraint_type, count(*) from dba_constraints where owner in ('PSPADM') group by constraint_type order
by 1;
select status, count(*) from dba_triggers where owner in ('PSPADM') group by status;
```

1.5 Final step for Goldengate replication from source(11g) to target (11g)

On Target DB server:

- set up supplemental log

```
SELECT SUPPLEMENTAL_LOG_DATA_MIN, FORCE_LOGGING FROM V$DATABASE;

ALTER DATABASE ADD SUPPLEMENTAL LOG DATA;
ALTER DATABASE FORCE LOGGING;
ALTER SYSTEM SWITCH LOGFILE;

SELECT SUPPLEMENTAL_LOG_DATA_MIN, FORCE_LOGGING FROM V\$DATABASE;
```

- Grant permission to GGT

```
grant create session to GGT;
grant DBA to GGT;
exec dbms_goldengate_auth.grant_admin_privilege ('GGT');
grant exempt access policy to GGT;

GRANT SELECT ANY DICTIONARY TO GGS;
```

- Create credentials and checkpoint. You need to replace <RDS SID> and <password> before running the command.

```
obey ./diroby/dblogin_t.oby
add checkpointtable
```

- Add schematranddata

```
obey ./diroby/dblogin_s.oby
delete schematrandata pspadm
add schematrandata pspadm
```

- Execute sequence.sql to support sequence replication

```
SQL> @sequence.sql
Please enter the name of a schema for the GoldenGate database objects:
GGS
Setting schema name to GGS

UPDATE_SEQUENCE STATUS:

Line/pos
----------------------------------------
Error
-----------------------------------------------------------------
No errors
No errors


GETSEQFLUSH

Line/pos
----------------------------------------
Error
-----------------------------------------------------------------
No errors
No errors


SEQTRACE

Line/pos
----------------------------------------
Error
-----------------------------------------------------------------
No errors
No errors


REPLICATE_SEQUENCE STATUS:

Line/pos
----------------------------------------
Error
-----------------------------------------------------------------
No errors
No errors


STATUS OF SEQUENCE SUPPORT
--------------------------------------------------------------
SUCCESSFUL installation of Oracle Sequence Replication support
```

- grant permissions to GGT

```
GRANT EXECUTE ON GGS.updateSequence to GGT;
GRANT EXECUTE ON GGS.replicateSequence to GGT;
```

- Setup and start the source to target replicat process on target db server

```
obey ./diroby/create_rpspuatq1.oby
info all
start replicat rpspuatq1 aftercsn <SCN from data export step>
start replicat rpspuatq1 aftercsn 4426865575

--> need to do below as we will be using rman restore and getting SCN after resetlogs
start replicat rpspuatq1, ATCSN <SCN from DB incomplete recovery>

info all
```

- **On the source side**, run the following GG commands to flush sequence

```
# If primary db is in QDC:
 obey ./diroby/dblogin_s.oby
# If primary db is in LVDC:
obey ./diroby/dblogin_s_remote.oby
 flush sequence PSPADM.*
```

- Wait until there is no replication lag showed up from "info all" command
- Create the Goldengate heartbeat table on source – we can skip this step as of now

```
# If primary db is in QDC:
cd /u01/ogg/scripts
sqlplus / as sysdba @monitor_gg_sync_table_ihp.sql

# If primary db is in LVDC
vi monitor_gg_sync_table_ihp.sql – changing "connect ops_user/&1" to "connect ops_user/&1@<Primary db name>"
sqlplus sys@<primary db name> as sysdba @monitor_gg_sync_table_ihp.sql
```

- On target, verify GG replication – we can skip this step as of now

```
SQL> alter session set nls_date_format='yyyy-mm-dd hh24:mi:ss';
Session altered.
SQL> select * from qbo.gg_heartbeat;
SOURCE               LAST_UPDATE
-------------------- -------------------
IHP                  2017-02-11 11:52:00

Make sure that the record IHP is close to current time (the difference should not be great than 1 minute)
```

- Run the following scrips to verify sequence replication

```
On Source and target :

set pagesize 1000
set numwidth 30
col SEQUENCE_OWNER format a10
col SEQUENCE_NAME format a32
select SEQUENCE_OWNER, SEQUENCE_NAME, LAST_NUMBER from dba_sequences where SEQUENCE_OWNER in ('PSPADM') order
by SEQUENCE_OWNER, SEQUENCE_NAME;
```

The output will looks like below and #s on target will be always not smaller than #s on source

1.6 Setup Veridata

- Folllow Setup veridata agent to setup veridata agent

1.7 Upgrade Oracle database on target to 12c

- Stop Goldengate on new-box for database 11g

```
. /l/<ORACLE_SID>
cd $GG_HOME
./ggsci
info all
stop rpspuatq1
info all
info rpspuatq1
NOTE: Make a note of the trail file number and RBA ->  we need this to start the new 12c replicat -> In this
case make a note of tr000000 and 37339444
obey ./diroby/dblogin_t.oby
dblogin USERIDALIAS ggtarget
delete rpspuatq1
```

- Follow Database Upgrade upgrade database to 12c on new-box
- At this point rename the profiles on new-box:

```
new-box:
mv /l/<oracle_sid> /l/<oracle_sid>_11g
mv /l/<oracle_sid>_12c /l/<oracle_sid>
```

- Create new password file for 12c:

```
new-box:
. /l/<oracle_sid>
cd $ORACLE_HOME/dbs
rm -f orapw<oracle_sid>
orapwd file=orapw<oracle_sid> entries=2
```

1.8 Setup Goldengate from 11g to 12c on new-box

1.8.1 set up GG for 12c on new/target db server

- Create GG sub directories for GG on oracle 12c

```
. /l/<oracle_sid>
cd $GG_HOME
./ggsci
create subdirs
exit
mkdir diroby dirdat
```

- Copy parameter, obey and other files for GG on oracle 12c:

```
dirprm        Copy these from 11g_gg_home/dirprm and change oracle_home in extract and replicat parameter files
for "SETENV" parameter.
diroby        Copy these from 11g_gg_home/diroby
dircrd        Copy from 11g_gg_home/dircrd
dirwlt        Copy from 11g_gg_home/dirwlt
```

- Set the new Golden Gate parameter to true for 12c

```
alter system set enable_goldengate_replication = TRUE scope=both;
```

1.8.2 Set up and start integrated replicat on 12c database

- At this point, the manager is running from the old home. Do not start the new manager at this point.
- Create the new replicat

```
obey ./diroby/create_rpspuatq1.oby
info all

. register the new replicat with the database
obey ./diroby/dblogin_t.oby
dblogin USERIDALIAS ggtarget
register replicat rpspuatq1 database

. alter the replicat and point it to its original position
alter replicat rpspuatq1, EXTSEQNO <trail num from 11g replicat>, EXTRBA <RBA from 11g replicat>
info rpspuatq1
```

- Stop the pump on the 11g db server

```
cd $GG_HOME
./ggsci
stop PPSPUTQ2
info all
```

- Take down the old manager(11g) on target db server

```
. /l/<oracle_sid>_11g
cd $GG_HOME
./ggsci
info all
stop manager
info all
exit
```

- copy/verify trail files are present on new 12c GG HOME dirdat location
- Start the new manager(12c) on the target db server

```
. /l/<oracle_sid>
cd $GG_HOME
./ggsci
start manager
info all
exit
```

- Start the pump on the 11g db server (source)

```
. /l/<oracle_sid>
cd $GG_HOME
info PPSPUTQ2
start PPSPUTQ2
info all
```

- Finally start the replicat on 12c gg home:

```
. /l/<oracle_sid>
cd $GG_HOME
start rpspuatq1
info all
```

- ### 1.9 Setup Goldengate replication from 12c to 11g

WARNING:

- **Perform this step only after Veridata comparison shows that the databases are sync.**
- **As soon as you start to perform this step, you should not perform any data fix or recreating schema or tables in the target because it will be replicated to the DB in source. Likely you have to clean up the target to source replication before you perform these tasks. Please always work with another DBA if you have to perform these tasks.**

- On 12c db server, create extract and pump

```
GGSCI (puatpspdb801.ie.intuit.net) 2> obey ./diroby/create_epspuatq2.oby

GGSCI (puatpspdb801.ie.intuit.net) 3> dblogin USERIDALIAS ggsource

Successfully logged into database.

GGSCI (puatpspdb801.ie.intuit.net as ggs@pspuat) 4> add extract epsputq2, TRANLOG, BEGIN NOW

EXTRACT added.


GGSCI (puatpspdb801.ie.intuit.net as ggs@pspuat) 5> add exttrail ./dirdat/pspuatq2/tr extract epsputq2,
MEGABYTES 500

EXTTRAIL added.


. register extract epsputq2 database

GGSCI (puatpspdb801.ie.intuit.net as ggs@pspuat) 7> obey ./diroby/create_ppspuatq1.oby

GGSCI (puatpspdb801.ie.intuit.net as ggs@pspuat) 8> add extract ppsputq1, EXTTRAILSOURCE ./dirdat/pspuatq2/tr

EXTRACT added.


GGSCI (puatpspdb801.ie.intuit.net as ggs@pspuat) 9> add rmttrail ./dirdat/pspuatq2/tr, extract ppsputq1,
MEGABYTES 500

RMTTRAIL added.

GGSCI (puatpspdb801.ie.intuit.net as ggs@pspuat) 10> info all

Program     Status      Group       Lag at Chkpt  Time Since Chkpt

MANAGER     RUNNING
EXTRACT     STOPPED     EPSPUTQ2    00:00:00      00:00:30
EXTRACT     STOPPED     PPSPUTQ1    00:00:00      00:00:09
REPLICAT    RUNNING     RPSPUTQ1    00:00:00      00:00:05
```

- On 12c, start extract and pump

```
GGSCI (puatpspdb801.ie.intuit.net as ggs@pspuat) 11> start EPSPUTQ2

Sending START request to MANAGER ...
EXTRACT EPSPUTQ2 starting

GGSCI (puatpspdb801.ie.intuit.net) 3> sh ls -ltr ./dirdat/pspuatq2

total 4
-rw-r----- 1 oracle dba 1467 Feb  3 02:02 tr000000000

GGSCI (puatpspdb801.ie.intuit.net as ggs@pspuat) 16> start PPSPUTQ1

Sending START request to MANAGER ...
EXTRACT PPSPUTQ1 starting


GGSCI (puatpspdb801.ie.intuit.net as ggs@pspuat) 17> info all

Program     Status      Group       Lag at Chkpt  Time Since Chkpt

MANAGER     RUNNING
EXTRACT     RUNNING     EPSPUTQ2    00:00:00      00:00:01
EXTRACT     RUNNING     PPSPUTQ1    00:00:00      00:01:18
REPLICAT    RUNNING     RPSPUTQ1    00:00:00      00:00:04
```

- On 11g, verify the trail file from 12c db server

```
oracle@puatpspdb800.ie.intuit.net:/u01/ogg/pspuat/12.2.0.2.2_12c->ls -ltr dirdat/pspuatq2/
total 0
-rw-r----- 1 oracle dba 0 Feb  3 02:03 tr000000000
```

- On 11g db server, create replicat

```
GGSCI (puatpspdb800.ie.intuit.net) 2> obey ./diroby/create_rpspuatq2.oby

GGSCI (puatpspdb800.ie.intuit.net) 3> dblogin USERIDALIAS ggtarget

Successfully logged into database.

GGSCI (puatpspdb800.ie.intuit.net as ggt@pspuat) 4> add replicat rpsputq2, INTEGRATED, EXTTRAIL ./dirdat
/pspuatq2/tr

REPLICAT (Integrated) added.


GGSCI (puatpspdb800.ie.intuit.net as ggt@pspuat) 5> dblogin USERIDALIAS ggtarget
Successfully logged into database.

Note: Do below step only if db version is 12c
GGSCI (puatpspdb800.ie.intuit.net as ggt@pspuat) 6> register replicat rpsputq2 database

2018-02-03 02:14:47  INFO    OGG-02528  REPLICAT RPSPUTQ2 successfully registered with database as inbound
server OGG$RPSPUTQ2.
```

- On 11g, Add checkpointtable and Start replicat

```
GGSCI (puatpspdb800.ie.intuit.net) 2> dblogin USERIDALIAS ggtarget
Successfully logged into database.

GGSCI (puatpspdb800.ie.intuit.net as ggt@pspuat) 3> add checkpointtable

No checkpoint table specified. Using GLOBALS specification (GGT.CHECKPOINT)...

Successfully created checkpoint table GGT.CHECKPOINT.

GGSCI (puatpspdb800.ie.intuit.net as ggt@pspuat) 5> start RPSPUTQ2

Sending START request to MANAGER ...
REPLICAT RPSPUTQ2 starting


GGSCI (puatpspdb800.ie.intuit.net as ggt@pspuat) 6> info all

Program     Status      Group       Lag at Chkpt  Time Since Chkpt

MANAGER     RUNNING
EXTRACT     RUNNING     EPSPUTQ1    00:00:03      00:00:06
EXTRACT     RUNNING     PPSPUTQ2    00:00:00      00:00:07
REPLICAT    RUNNING     RPSPUTQ2    00:00:00      00:00:00
```

- On 12c, run the following GG commands to flush sequences

```
GGSCI (puatpspdb801.ie.intuit.net) 2> obey ./diroby/dblogin_s.oby

GGSCI (puatpspdb801.ie.intuit.net) 3> dblogin USERIDALIAS ggsource

Successfully logged into database.

GGSCI (puatpspdb801.ie.intuit.net as ggs@pspuat) 4> flush sequence pspadm.*
Successfully flushed 69 sequence(s) pspadm.*.
```

- Verify sequence bi-directory replication

```
1. On source, create test sequence if it does not exists:

SQL> create sequence pspadm.test;


2. Run the following SQL on 11g and 12c to confirm the sequence exists

SQL> select SEQUENCE_OWNER, SEQUENCE_NAME, LAST_NUMBER from dba_sequences where SEQUENCE_OWNER = 'PSPADM' and
SEQUENCE_NAME = 'TEST';


3. Test the sequecne

a) Run it on 11g:
SQL> select pspadm.test.nextval from dual;

   NEXTVAL
----------
       1

b) Wait 10 seconds. Run it on 12c
SQL> select pspadm.test.nextval from dual;

   NEXTVAL
----------
       22


Note: the # here should be higher than one in step above. If not, replica this step until this is true. If this
is not case after trying for a couple of minutes

c) Wait 10 seconds. Run it on 11g
SQL> select pspadm.test.nextval from dual;

   NEXTVAL
----------
       43


Note: the # here should be higher than one in step above. If not, replica this step until this is true. If this
is not case after trying for a couple of minutes
```

- ## 1.10 Setup GG monitoring on source and target db server

- On target, add new extract and pump to monitor_pspuat.cfg

```
[oracle@pprdc34uw214g238195 scripts]$ cd /scripts/ogg


Edit the new file based on which processes need to be monitored.
oracle@puatpspdb801.ie.intuit.net:/u01/ogg/pspuat/12.2.0.2.2_11g->cat /scripts/ogg/monitor_pspuat.cfg
MANAGER,1
EPSPUTQ2,2
PPSPUTQ1,2
RPSPUTQ1,2


NOTE:
The monitor configuration file (monitor_qboqpd34.cfg),
has one entry for the manager and one entry for each of the processes (Extract,
pump or replicat). The number after each entry separated by a comma, is simply
a flag:
0 = do not monitor this process
1 = monitor this process
2 = monitor this process and additionally report on
the lag and checkpoint
```

- On IHP, create the file called login under /u01/ogg/scripts/ . – skip for now

```
cd /u01/ogg/scripts
vi login
cat login
ops_user/"<real password here>"
chmod 700 login
```

- On both sides, make sure to have the following crontab entries

```
##Goldengate monitoring##
*/5 * * * * /scripts/ogg/monitor_ggserr.sh pspuat 1>/dev/null 2>&1
58 23 * * * /scripts/ogg/rotate_ggserr.sh pspuat 1>/dev/null 2>&1
#
```

2. Cutover to 12c

2.1 Check OOS (Out-Of-Sync)

Verify latest veridata compare run to make sure there are NO OOS rows, if its there we need to fix it first before cutover. Below is example :

```
1. log in to 12c db as GGT user

2. check OOS rows in veridata for table and OOS rows and columns

3. run below update on target(12c) database as GGT user. Change table name as per OOS report.
update pspadm.PSP_ADDRESS set (ADDRESS_LINE2)=(select ADDRESS_LINE2 from pspadm.PSP_ADDRESS@to_source where
ADDRESS_SEQ='bd8ae74e-d3a4-4fb9-a2c4-eb0cf9c2c7b6' and REALM_ID='-1')
where ADDRESS_SEQ='bd8ae74e-d3a4-4fb9-a2c4-eb0cf9c2c7b6' and REALM_ID='-1';
```

2.2 shut down application on current 11g

make sure all connections are gone

```
set lines 300
col service_name for a30
col machine form a40;
select username,machine,count(*) from gv$session where username is not NULL and username not in
('SYS','SYSTEM') group by service_name,username,machine;

--> make sure there are connections from %PSP% user.
USERNAME                       MACHINE                                    COUNT(*)
------------------------------ ---------------------------------------- ----------
DBSNMP                         oprdoemas303.corp.intuit.net                      2
PUBLIC                         pprdpspdb801.ie.intuit.net                        1
DBSNMP                         oprdoemas302.corp.intuit.net                      5
GGS                            pprdpspdb901.ie.intuit.net                        5
GGT                            pprdpspdb901.ie.intuit.net                        1
DBSNMP                         pprdpspdb901.ie.intuit.net                        4

6 rows selected.

-- kill app connections (if required after shutting down application)
select 'ALTER SYSTEM KILL SESSION  '''  || SID || ',' || SERIAL# ||  ''' IMMEDIATE ;' from V$SESSION where
USERNAME='PSP%';
```

2.3 Lock application user accounts on current database (11g)

```
alter user PSPADM account lock;
alter user PSPAPP account lock;
alter user PSPBOADMIN account lock;
alter user PSPFLUXADM account lock;
alter user PSPLOG account lock;
alter user PSPFLUXADMIN08 account lock;
alter user PSPLOG  account lock;
alter user PSPREAD  account lock;
alter user PSPTEMP  account lock;
```

2.4 Verify GG replication

```
GGSCI (pprdpspdb900.ie.intuit.net) 1> info all

Program     Status      Group       Lag at Chkpt  Time Since Chkpt

MANAGER     RUNNING
EXTRACT     RUNNING     EPSPLPD1    00:00:00      00:00:04
EXTRACT     RUNNING     PPSPLPD2    00:00:01      00:00:02
REPLICAT    RUNNING     RPSPLPD2    00:00:00      00:00:07

GGSCI (pprdpspdb901.ie.intuit.net) 1> info all

Program     Status      Group       Lag at Chkpt  Time Since Chkpt

MANAGER     RUNNING
EXTRACT     RUNNING     EPSPLPD2    00:00:04      00:00:06
EXTRACT     RUNNING     PPSPLPD1    00:00:00      00:00:06
REPLICAT    RUNNING     RPSPLPD1    00:00:00      00:00:07
```

2.5 Unlock application user accounts on current database (12c)

```
alter user PSPADM account unlock;
alter user PSPAPP account unlock;
alter user PSPBOADMIN account unlock;
alter user PSPFLUXADM account unlock;
alter user PSPLOG account unlock;
alter user PSPFLUXADMIN08 account unlock;
alter user PSPLOG  account unlock;
alter user PSPREAD  account unlock;
alter user PSPTEMP  account unlock;
```

2.5 start application on new database 12c

```
set lines 300
col service_name for a30
col machine form a40;
select username,machine,count(*) from gv$session where username is not NULL and username not in
('SYS','SYSTEM') group by service_name,username,machine;
```